Engine Model(s): PT6A

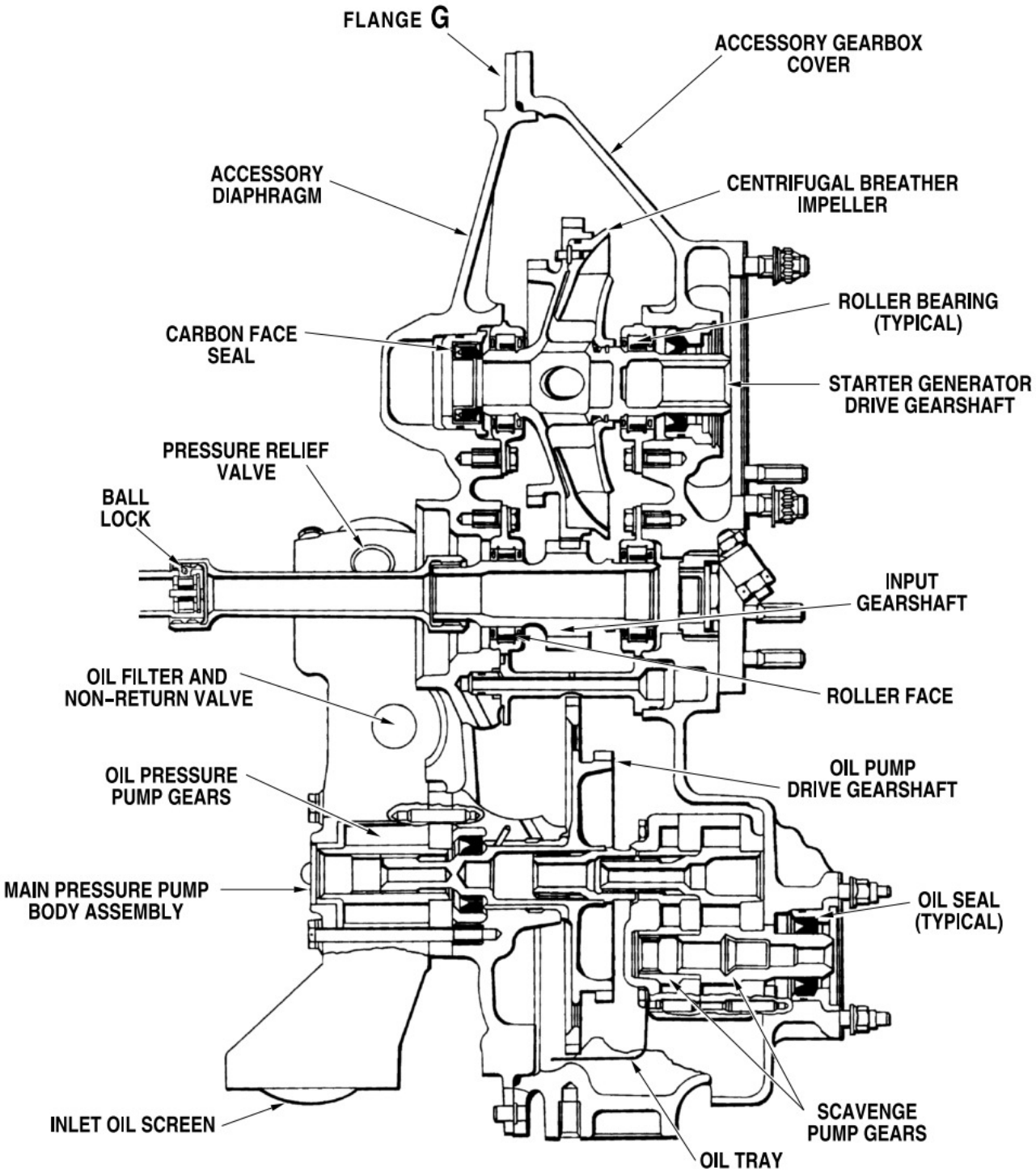
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ACCESSORY GEARBOX ASSEMBLY - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Accessory Gearbox (Typical Cross-section)



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The accessory gearbox, located at the rear of the engine, consists of two magnesium alloy castings, bolted together at their outer flanges and secured to studs on Flange G of the compressor inlet case (Ref. 72-20-00).

The front casting, provided with preformed packings at the front and rear mating faces, forms an oil-tight diaphragm between the oil tank compartment in the compressor inlet case and the accessory gearing. The diaphragm also provides support for the oil pump and pressure relief valve assembly, which is secured on the front face by four bolts, and the accessory drive gear front bearings and seals. The center of the diaphragm is sealed from the oil tank by the inlet case center tube. The diaphragm is mounted on the gearbox rear housing by four countersunk head screws and self-locking nuts located at the 4th, 8th, 14th and 18th positions in clockwise rotation, assuming the 1st to be at the 12 o'clock position.

The rear casting forms the gearbox housing and provides support bosses for the accessory drive gear rear bearings and seals. Roller bearings secured to the gearbox rear casting and the diaphragm at the center provide support for the gearbox input gearshaft. The input gearshaft incorporates an external spline at its front end for mating with a corresponding internal spline on the input coupling shaft. The coupling shaft is secured at its front end to the compressor rear hub coupling spline; positive engagement is obtained by a ball-lock in the spline coupling arrangement, which transmits thrust to the compressor No. 1 ball bearing.

The internal scavenge oil pump, consisting of front and rear elements, is mounted at the bottom of the gearbox compartment and driven by the No. 3 drive. Each element consists of twin gears operating in a ported housing; the front element scavenges the No. 2 bearing area and dumps the oil into the gearbox sump. The rear element scavenges the gearbox sump and returns the oil to the airframe-mounted oil cooler and then to the tank.

The external scavenge oil pump, consisting of front and rear elements, is mounted on an external pad on the gearbox housing and driven by the No. 4 drive. The pump is similar to the internal pump; however, the housings of each element incorporate large input elbows which accommodate their respective external scavenge tubes. The front element scavenges the No. 3 and 4 bearing area and dumps the oil into the gearbox sump, while the rear element scavenges the reduction gearbox and directs the oil through an airframe-mounted cooler, and then to tank.

All main accessory drives are supported on identical roller bearings. With the exception of the front bearing of the starter-generator gearshaft, all drives are fitted with garter-type oil seals. An oil tray is incorporated in the lower section of the gearbox adjacent to the scavenge pump drive gears (drives 3 and 4), to minimize oil foaming during engine operation. A centrifugal oil separator, installed on the starter-generator gearshaft, separates oil from engine breather air and discharges relatively oil free air to atmosphere via cored passages in the gearbox diaphragm. Oil drains into the gearbox sump. A carbon-face seal fitted at the front end of the gearshaft and within the cored breather air passage prevents oil leakage through the bearing assembly. The starter-generator gearshaft may be of dry spline Pre-SB1386 or wet spline Post-SB1386 configuration.

Mounting pads, complete with studs, are incorporated on the external face of the gearbox housing for the fuel control unit/fuel pump, airframe-supplied starter-generator and Ng tachometer-generator. Two additional pads are incorporated on the left side of the housing and are available for additional bolt-on airframe-supplied equipment. The drive gears at these pads are optional

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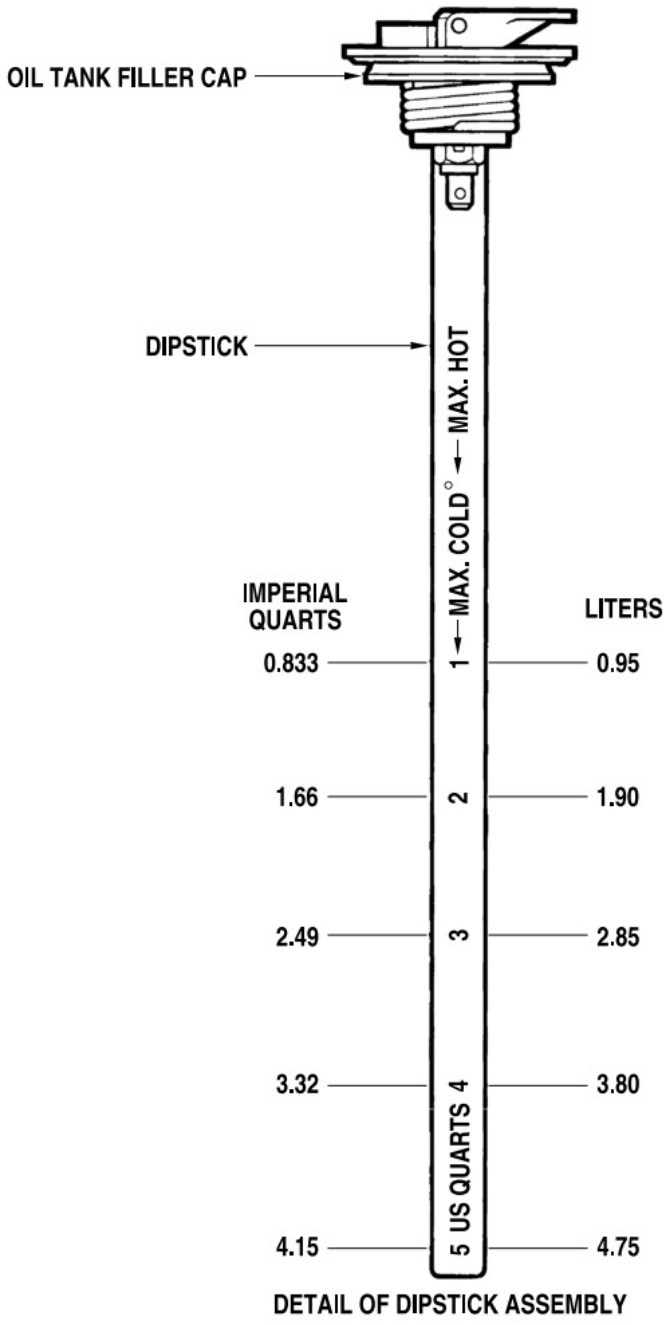
equipment. A mount pad, complete with studs, is also provided at the rear face of the external scavenge pumps for airframe-supplied equipment. The drive gear is a customer-option. An access plug, located below the starter-generator pad, provides access for a puller tool which must be used to disengage the ball-locked input gearshaft and compressor rear hub coupling shaft during disassembly.

Engine oil pressure is regulated by a plunger-type pressure relief valve located and secured to a boss at the top of the oil pump housing. All oil in excess of the regulated pressure is returned directly into the tank. The relief valve consists of a valve sleeve, valve, spacers, spring and spring seat, guide and retaining ring, which are retained within the valve housing by a second retaining ring. The assembly is accessible through the oil filter boss on the compressor inlet case, with the filter and filter housing removed. The pressure relief valve is adjusted during engine assembly, to provide the required oil pressure, by insertion of a suitable number of spacers under the spring. Under normal conditions, this setting will require no further adjustment.

A combined filler cap and dipstick/gage assembly (Ref. Fig. 2) is installed at the 11 o'clock position of the accessory gearbox housing. The assembly is secured to a removable filler tube and housing by a quick-release and locking arrangement. The filler tube projects forward and downward through the accessory gearbox and diaphragm and into the oil tank to form a seal between the tank and gearbox. The dipstick blade is calibrated in U.S. quarts and indicates the quantity required to fill the tank to maximum oil level for both hot and cold oil conditions.

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Figure 2 Dipstick and Filler Cap



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